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Next item →

1. In the context of Keras, what is the primary purpose of a transpose convolution layer in a model? 1 / 1 point

- ☐ To combine multiple input feature maps into a single output feature map.
 - ☐ To reduce the computational complexity of the model.
 - ☒ To increase the spatial resolution of the input image.
 - ☐ To perform max pooling operations.
- ✔ **Correct**
Correct! The transpose convolution layer, also known as deconvolution, is primarily used to increase the spatial resolution of the input image.

2. In the context of a fully connected GAN generator, which of the following layers are typically included? 1 / 1 point

- ☐ Pooling layer
 - ☒ Hidden layers
- ✔ **Correct**
Correct! Hidden layers are used to transform the input data into a meaningful output.
- ☒ Input layer
- ✔ **Correct**
Correct! The input layer is where the random noise vector is fed into the generator.
- ☐ Normalization layer
 - ☒ Output layer
- ✔ **Correct**
Correct! The output layer generates the final fake image.

3. Select all components that are typically part of the discriminator model in a GAN. 0.9 / 1 point

- ☒ Dropout layers
- ✔ **Correct**
Correct! Dropout layers are used to prevent overfitting in the discriminator model.
- ☒ Convolutional layers
- ✔ **Correct**
Correct! Convolutional layers are a fundamental part of the discriminator model.
- ☐ Pooling layers
 - ☒ Activation functions
- ✔ **Correct**
Correct! Activation functions are used to introduce non-linearity in the model.
- ☐ Embedding layers
 - ☒ Batch normalization layers
- ✔ **Correct**
Well done! Batch normalization layers are used to stabilize and accelerate training.
- ☐ Recurrent layers
- You didn't select all the correct answers

4. Which method is used to train a model with a batch of images or data in GANs? 1 / 1 point

- ☐ compile_batch
 - ☒ train_on_batch
 - ☐ fit_on_batch
 - ☐ evaluate_batch
- ✔ **Correct**
Correct! The train_on_batch method is used to train a model with a batch of data.

5. Which of the following are key aspects of defining a generator function for a DCGAN? 1 / 1 point

- ☐ Applying L2 regularization to weights
 - ☒ Incorporating batch normalization
- ✔ **Correct**
Correct! Batch normalization helps in stabilizing the training process.
- ☒ Utilizing transposed convolutional layers
- ✔ **Correct**
Correct! Transposed convolutional layers are essential for upsampling in the generator.
- ☐ Adding max pooling layers
 - ☒ Using a sigmoid activation function in the output layer
- ✔ **Correct**
Correct! The sigmoid activation function is often used in the output layer to produce pixel values between 0 and 1.